



SECURE NETIZENS

C4E1J5

Learning Outcome

After this lesson, learners will be able to:

- Understand the different types of digital footprints and their implications
- Know how to remain safe on social media
- Use browser safely to access the internet
- Set strong password to keep digital devices safe
- Keep online accounts safe using Two-Factor authentication
- Keep device safe while accessing internet the through Wi-Fi
- Understand the WhatsApp threats and preventions

With the extensive use of the internet and networking everywhere, it is the foremost requirement to know how to connect carefully before stepping into the virtual world of the internet so that our safety and security can be ensured in this virtual world. A personal computer used without proper security measures could also lead to exploiting the system for illegal activities.

3.1 DIGITAL FOOTPRINTS

With the aid of high-speed Internet, an increasing number of individuals are adopting digital technology nowadays. Whenever we surf the web, knowingly or unknowingly, using smartphones, tablets, computers, etc., we leave a trail of data reflecting the

activities performed by us online. This irreversible digital trail is known as a "**digital footprint**" or "digital shadow."



Users on the web are using the technology very frequently without knowing the facts about digital footprints. Digital footprints are stored permanently. Every time you do something online, like gaming, sending messages, making online reservations, etc., the trail is stored forever. This digital trail has been further categorised as active and passive footprints.

Active digital footprints involve the data that you intentionally submit online. This would include emails we write or responses or posts we make on different websites or mobile apps, etc.

Passive digital footprints involve the data that you unintentionally submit online. Here, users don't exchange data, but a trail of their activities is left behind and can be used to identify them later. Website visits and actions, searches and online purchases, online reviews, and feedback are among the activities that add passive data traces to your digital footprint. To understand user sentiments, marketing trends, or statistics, this data may occasionally be used.

Case Study



Ranvir is a frequent user of Facebook, Twitter, Instagram, etc. to stay in touch with his loved ones. He has numerous Facebook friends. He used to post his every activity on Facebook and wait for the comments. He also replies to the comments very frequently. This implies that he is very active on the web and that he is intentionally establishing his active digital footprints on the internet.

** Ranvir could be marked as a target as a result of active digital footprints.

Impact of Digital footprints on Ranvir's life

1. Someone on social media can create wrong opinions about Ranveer and it can spoil his social relations.
2. Ranveer's every activity is public, so he can be a victim of stalking or blackmail.
3. Photographs uploaded on social media platforms get lost in the cloud. They can later be misused through photo morphing.
4. Employers may search online for Ranvir's social media status in the future and get a messy set of digital footprints.

Case Study



Kiyra, a 14-year-old girl, downloads the car game application onto her tablet.

While downloading that online gaming app, without reading the pop-up messages, she unknowingly allowed the apps to access her current geolocation. She started playing with unknown online players without knowing the consequences. In this scenario, she left passive digital footprints.

Impact of Passive Digital footprints on Kiyra's life

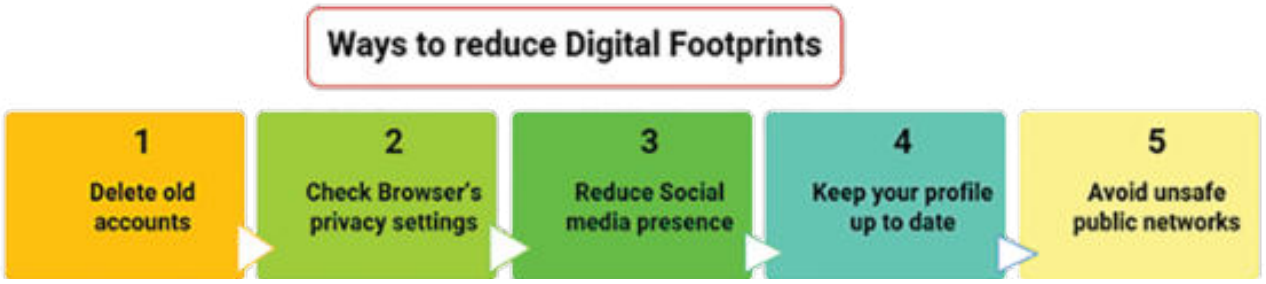
1. Everyone knows where she is because her geolocation is public on the internet.
2. She may have started getting requests for similar games because advertisers compile and analyse her interests and provide targeted advertisements.

Digital footprints are not always bad. They can also improve your reputation and increase the occurrence of business opportunities.

Resources:

Digital Footprints	https://www.youtube.com/watch?v=dmQGq_FNBpE
Internet Danger: Everyone knows Sarah	https://youtu.be/ThxmgXMBpoM

Safety measures to reduce the digital footprints are as follows:



MEDIA REPORTS

MEDIA REPORTS	QR CODE
Digitl footprints helps the cyber cell to investigate **Shraddha Walkar murder: Digital footprints most important to nail Aftab in court, say Mumbai cyber crime expert**	

Positive impact of digital footprints



3.2 BROWSER SECURITY

What is a browser?



For connecting with the digital world, we need an electronic device like a smartphone /tablet / PC/laptop and an internet connection. Once we are connected to the Internet, we can access websites using an application called a web browser. Web browser is an application that is used by a user to access the internet. There are many different web browsers available but none of them are totally safe from cyber security threats.

Securing Your Web Browser

The web browsers are vulnerable to security threats, and interacting with online content invites the risk of encountering viruses, malware and other threats. Unblocked popups, malicious browser extensions, unsafe plugins, weak passwords, the browser's password saving feature etc. are some of the most commonly exploited weaknesses of a web browser. To assure online protection the security of browsers is a mandate step. Many users do not know how to configure their web browsers securely. It is important to enable security features in the web browser to minimize the risk to the computer.

Case Study





1. Mr. Paresh, a Mathematics teacher at Bal Sunder School, has opened his email ID in a browser in the computer lab during his free period to check a very important mail. After the task, he logged out of his email id safely. Immediately after this, to his surprise, he found that someone had hacked his account and his contacts had started getting embarrassing emails from his ID. On enquiry, it was found that someone from a very distant location had acquired his account details through the browser which he had used to check his email. This happened because the browser used by Mr Paresh saved his email login credentials.
2. Meeta used to visit abcMusic.com to listen to online music albums. One day, while listening to music she came across a pop-up displaying a link to download the latest songs for free. When she clicked the link, she was asked to share some of her personal identifiable information. She filled the form without giving a second thought. One month later, she was frightened to find some unauthorised transactions in her credit card statement.

Security Extensions in Browsers:

There are many browser extensions or plug-ins available to safeguard your browser. Some of them are listed as follows:

Web of Trust (All Browsers)

Web of Trust is a plug-in supported by almost all web browsers that informs you whether a website you are visiting is trustworthy or not. That way if you happen to visit a website that is not secure, you get a warning message.

LastPass (All Platforms)

Last Pass is a browser plugin that helps to generate a strong password and stores it in an encrypted form. It helps you to manage your passwords.

Note: Add the browser extensions or plugins only through the browser's extensions. Understand the features and the risks involved before installing the extensions or plugins.

To add extensions/plugins to the Chrome browser, use the following steps:

1. Click on three dots present on the extreme top right side of the browser.
2. Now click on more tools and then click on extensions.



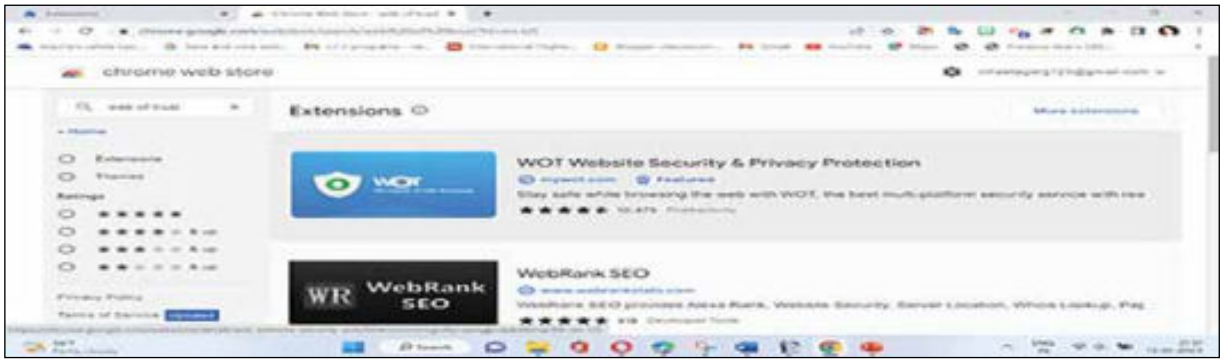
3. Click on three horizontal lines present on the top left side.



4. Now, click on Open Chrome Web store.



5. Search the desired extension/plugin and install it.



Safety Measures

1. Configure browser's "Privacy and Security" settings.
2. Enable pop-up blocking.
3. Do not click on malicious links in an email or message instead type the URL into the address bar of your browser.
4. Always use a secured web browser to avoid the risks.
5. Keep your web browser fully patched and up to date.
6. Install a personal firewall along with anti-virus software with the latest virus signatures that can detect malware such as keyloggers.
7. Regularly change your passwords with the combinations of letters, numbers and special case characters in critical web applications if a one-time password system is not supported.
8. Turn off all JavaScript or ActiveX support in your web browser before you visit any unknown websites.
9. Delete browsing history which deletes all the cookies, temp files, history, active X filtering etc.

Supportive Resources

Safe Web Browsing	https://www.youtube.com/watch?v=Q_mqjS_E9mg
Internet Safety: Your Browser's Security Features	https://youtu.be/2ZZQlgV2Gus
Web Of Trust	https://youtu.be/PQl1d5jFRnI

3.3 SECURE YOUR DIGITAL DEVICES

Securing your devices is crucial in protecting your personal and sensitive information. Devices such as smartphones, laptops, and tablets store a vast amount of personal information, including emails, contacts, financial information, and sensitive files. Unauthorized access to your devices can compromise the privacy of your data, files, and online accounts. With access to your device, hackers could easily steal your login credentials and gain access to your email, social media, or banking accounts.

Passwords are a critical aspect of securing devices. Without a strong and unique password, anyone can easily gain access to your device, sensitive information, and online accounts.

Case Study



Ms. Sanya, a class X student, had just created her profile on a popular social networking site with the help of her friends. Although she decided the password on her own, others in the group were aware about her choice and confidential data which she had filled while creating the profile. Very soon she had been informed by one of her distant relatives that wrong messages and pictures were being sent through her profile. To her horror, when she tried to login to check the reality, she was denied access to her own profile because someone had already changed the password. Seeing her so depressed her ICT teacher enquired about the matter and on knowing about the incident, she guided her on how to lodge a complaint with the Cyber Police and got it solved.

3.3.1 Password Protection

A password has the same significance as that of a key in lock-key protected material. We can ensure the safety and security of our material and even of our identity in the cyber world with the help of secure passwords.

A lock can be opened and all the material kept safely inside the lock-key is protected box/room will be accessible by any person possessing the key in an authorized/ unauthorized way. Similarly, if accidently our password in the cyber world is leaked or compromised then our safety, security and identity will be at stake. Thus, it's very important to keep a strong password so that no unauthorized person can break it.

A password having

- minimum 8 alphanumeric characters
- combination of capital and small alphabets, digits and special symbols
- is considered a strong password.







Things to avoid when choosing a password

Some of the specific things which you should avoid while choosing a password are as follows:

- **Names of any kind.** These include your username, your first or last name in any form, or your spouse's or child's name.
- **Any kind of personal information.** This includes your address, birthdays, licence plate numbers, telephone numbers, etc.
- **Sensitive information.** This includes your ATM PIN, student or Job ID, Adhar Card number, or Credit Card number.
- **Words contained in English or foreign language dictionaries.** This includes obvious words like "password" or "safe" or "secret" etc.

Password-Passphrases

If you find it difficult to remember hard to crack passwords then the password passphrase method can be used. In this method, the user thinks of any phrase/sentence of his/her choice and uses the first letter of every word to create a password. It can be tweaked a bit so as to make a password that is hard to guess but easy to remember.

Look at the following examples:

- **Passphrase:** They like basking in sun in winter season
- **Password:** TlBi$1W$0
- **Passphrase:** The road to success is always under construction!
- **Password:** Trt$iauc!

3.3.2 Check Password Strength



If you are not sure whether your password is strong enough or not you can check the strength of your password by visiting a web site:

https://howsecureismypassword.net/

Note: After checking the strength of the password it is advisable not to keep the same password but a password similar to the tested password.

3.3.3 Techniques used by hackers/crackers to retrieve your passwords

Techniques used by hackers/crackers to retrieve your passwords

Shoulder Surfing



In this technique a person stands behind you and looks over your shoulders to steal passwords and other sensitive information.

This can be avoided by remaining extra vigilant and careful when entering your password especially at public places like Internet café. Do not reveal your passwords in front of others and do not type usernames and passwords before unauthorized persons. You can also cover the keyboard with paper or hand or something else to prevent them from being viewed by unauthorized users.

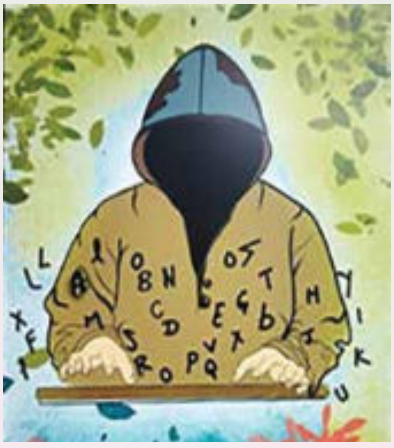
Brute force attacks



This is another way of stealing the password in which cyber thieves try to guess your password. They try all the possible combinations with the help of personal information of an individual like the person's name or nick name, pet's name, date of birth, phone numbers, etc. This is known as a "Brute force attack".

It is advisable not to use passwords that represent your personal identifiable information like name, address, phone number, etc.

Dictionary attacks



In this technique hackers try all possible combination of dictionary words to crack your password. This is called a "dictionary attack".

It is advised not to use dictionary words while creating the passwords for login accounts.

Supportive Resources

https://www.youtube.com/watch?v=aEmF3IylvR4

Safety Measures

- Don't disclose your password to anyone, including your partner, friends and family members.
- Keep your password hard to guess by making it a combination of alphanumeric characters, special characters and a mix of small and capital letters.
- Make a healthy habit of changing the passwords regularly like a toothbrush.
- Don't keep the same passwords for multiple logins and accounts.
- Don't write your password anywhere except in your own mind.
- Don't tell your password to someone on the phone or through mail or message.
- Be cautious while entering your password, one might be secretly eyeing it through your shoulder.
- Always do the password entry for financial transactions with the help of a virtual keyboard as it will help to avoid keylogging.

3.4 SECURE YOUR ONLINE ACCOUNTS

Thanks to technology, we can easily connect with people across the world, shop online for the goods of our choice, perform online financial transactions online and so on. However, securing your online accounts is crucial to safeguarding your personal and sensitive information, maintaining your privacy, avoiding data loss, and preventing financial losses. Unsafe online accounts could lead to identity theft or financial fraud. As discussed in the previous section, a strong password is a mandate to secure your online accounts. To further keep your online accounts (social media accounts, financial accounts etc.) safe and secure from cyber fraudsters besides password a two Factor authentication method can be used.

Two-Factor Authentication

As the name suggests, two-factor authentication requires one extra step —to log onto a site or access an online account. To get access to a website, you usually enter your username followed by a password. The site typically then sends a text message to your mobile phone with a numerical code. This code is called a verification code/ passcode/ authenticator. You can only access the site by entering the code that appears on your mobile device. In the absence of code, you can't access the website even if you have the correct password.

2FA offers an extra level/layer of security that cyberthieves can't easily access, because the criminal needs more than just your username and password credentials.



Supportive Resources

https://www.youtube.com/watch?v=AMOtB7XkTT4

Safety Measures

- There are different methods of 2FA such as email, SMS, and app-based authentication. Use a reliable 2FA method.
- Never reuse 2FA codes. Each code is unique and meant for a one-time use.

  When setting up 2FA, you will be given a set of backup codes. Keep these codes in a safe place in case you lose your device or cannot access your authentication app.
- Enable 2FA for all accounts that offer this option, including social media, email, banking and other important accounts.
- Keep an eye on your accounts and review your login history regularly. Change your password in case you notice any suspicious activity.

3.5 WI-FI SECURITY

Wi-Fi devices are widely used to access the Internet. But most of these devices are unsafe to use in their default configuration mode. Due to unawareness innocent people use these devices and get victimised. It

is therefore important that proper settings are made before actually using them so as to access the internet safely and securely.

3.5.1 Securing Home Wi-Fi

With the increased use of technology everywhere, it's very common to have multiple gadgets or devices at home connected to the internet through Wi-Fi. Home Wi-Fi is also vulnerable to very serious cyber threats so it's always advisable to protect home Wi-Fi with a strong password.

Case Study

Mr. Avijeet, a doctor by profession, had just got his dispensary cum home computerised with Wi-Fi enabled networking. He thought that as it's his personal network so why to keep it password protected as he wanted to have a hassle-free networking environment. To his surprise, he had observed that soon his internet consumption had increased many folds while he had only limited use. Before he could take it more seriously, he had served a notice of investigation by the Cyber Crime Branch that a serious cybercrime happened through his network. On detailed investigation, he got a clean chit on his involvement by the cyber cell and it was established that someone had misused his open Wi-Fi network to commit the crime.



Supportive Resources

https://www.youtube.com/watch?v=YiVkbOggOBQ

Safety Measures

1. Always keep your home Wi-Fi network password protected.
2. Always access password protected Wi-Fi networks.
3. Don't keep your Wi-Fi network in "auto-connect" mode.
4. Disconnect the Wi-Fi network if you are not using it.
5. Ensure that you visit the Sites with https rather than http in the URL.
6. Disable the automatic file sharing option.
7. All Wi-Fi equipment supports some form of encryption. So, enable them.
8. Enable MAC address filtering on Wi-Fi devices.
9. Avoid dynamic IP addresses for home Wi-Fi rather use static IP addresses.

3.5.2 Securing Public/Open Wi-Fi



Public Wi-Fi is a wireless network that is provided for public use in places such as airports, coffee shops, libraries, hotels, and other public spaces. It is usually free of charge and doesn't require any authentication or password to access the internet. It is a convenient way for people to access the internet when they are on-the-go or away from home.

More the openness of any network, the more it will be vulnerable to any type of attack. The Same is true in the case of Wi-Fi. One should be very cautious while accessing any public/open Wi-Fi as it is more vulnerable for eavesdropping i.e., unauthorised accessing of data when it's on the way on the network. Not only the data of the owner of the open Wi-Fi network will be at risk but anyone can misuse this open network to commit any type of crime. More than this, even the open Wi-Fi network can be used by hackers to distribute malware. So, in the interest of everyone, one should access only protected Wi-Fi networks.

Case Study



Ms. Ruby, an event designer in a multinational company, had a very hectic schedule in the month of May. She had lots of bill payments etc. due at her end. In the afternoon, after finishing her lunch she started paying her few of the bills online by credit card using a free Wi-Fi network available in the restaurant. To her horror, the moment she reached back to her seat to resume the work, she started getting alert messages of huge bills of shopping done through her credit card. Initially she could not understand how any shopping could be done through her credit card if she had that card but later on her friend in the IT section explained to her that someone might have hacked her credit card information the moment she used public Wi-Fi and misused it. Immediately, she lodged a complaint with her bank and got her credit card blocked.

Some real incidents that took place in recent years.

1. Hackers penetrated an open Wi-Fi network of luxury hotels owned by the Thompson Group in New York, Los Angeles and Washington DC and stole the private emails sent by the guests. The hackers then attempted to extort money from the hotel chain by threatening to publish the emails.
2. Just 5 minutes before Delhi blasts in September 2008, terrorists used an unsecured Wi-Fi connection of a company at Chembur in Mumbai to send terror emails to authorities and news channels. These hackers do not leave a trail of footprints for the investigators to arrive at a logical conclusion. The audit trail ends at the Wi-Fi Access Point of the legal user. So it becomes imperative for the users to secure their own Access Points (wireless router).

   Source: https://www.infosecawareness.in/concept/wifi-security?lang=en

Supportive Resources

https://www.youtube.com/watch?v=XcghUy-8VRA

Safety Measures

1. Always verify your Wi-Fi connection before connecting.
2. Ask the correct Wi-Fi connection name of hotel/mall etc. from the authorized person before connecting as some fake WiFi connections might be available to trap any user.
3. Always access password protected Wi-Fi networks.
4. Avoid checking sensitive data on public Wi-Fi networks.
5. Turn off automatic file sharing option.
6. Use a VPN
7. Turn off the your Wi-Fi if not in use.
8. Configure the wireless settings on your devices to not automatically connect to available Wi-Fi hotspots. This ensures that you do not connect to public networks unknowingly. You can do this by turning off the "Connect Automatically" feature on your device so that they do not auto-connect and search for known Wi-Fi networks.

3.6 WHATSAPP SECURITY

WhatsApp is safe and secure due to the end-to-end encryption technology used for messaging but you have to be more cautious. According to a survey, WhatsApp

crossed 200 million monthly active users in India in February 2017. You all know the benefits of WhatsApp but it is essential to know the safe use of WhatsApp. Generally, people **install fake WhatsApp applications that lead to privacy breaches**.

Procedure to identify the fake app

1. Unusual and unique app & developer name: Verify the name of the app & the developers.
2. Check the download count.
3. See the description of the application.
4. Screenshots & reviews: See What People Are Saying.

MEDIA REPORTS

Case Study

Installing Apps from unverified sources



Ramesh, a teenage boy, got a smartphone on his birthday. He was fond of playing mobile games but he lacked knowledge of how to download Apps.

Most of the time, he downloads the mobile gaming apps from unverified sources, like downloading a link sent by a friend, rather than from the verified Google Play Store.

One day, his phone was affected by harmful viruses and all his personal data got hacked by hackers.

Threats for WhatsApp users



1. Mobile phones can be hacked through malware scripts.
2. GIF images from unknown sources can harm the Operating System and applications.
3. There may be a chance of losing the personal information.
4. Unverified apps may drain out the phone memory.
5. Fake apps may mislead the objective.
6. Improper privacy settings open doors to threats.

Safety Measures (Keep your mobile safe)

1. WhatsApp is the favourite medium of hackers through which they can access your private information and transmit it to remote servers. To protect your valuable information, follow the guidelines below:
2. Keep your softwares up-to-date.
3. Don't download apps that look untrusted or shady.
    - Adjust your privacy settings to control who sees your information. You can set Last Seen & Online, Profile Picture, About, or Status to the following options:
    - Everyone: All users can see your profile photo, about, or status.
    - My Contacts: Only contacts in your address book can see your last viewed online profile picture, profile picture, and status.
    - My Contacts Except...: Except for excluded contacts, contacts in your address book can see your last online, profile photo, summary, or status.
    - Nobody: No one can see your last viewed or online content, your profile picture, overview or status.

Scan the QR code to learn about these privacy settings

ASSESSMENT

I. Multiple choice Questions

1. How can you hide your status update from being shared with every one of your contacts in whatsApp?

   A. By not updating the status page

   B. By using "Status privacy" option

   C. By using "Account" option

   D. None of the above

Ans B. By using "Status privacy" option

2. Two Blue Ticks on Message in WhatsApp means

   A. Message is sent

   B. Message sending failed

   C. Message read by the recipient

   D. Message is deleted

Ans. C. Message read by the recipient

3. Choose the correct way to keep your messages safe:

   A. Don't download apps that look untrusted

   B. Never block the unknown or Spam Numbers.

   C. Turn off all Security Notifications.

   D. None of these

Ans. A. Don't download apps that look untrusted

4. Which of the following is not true for browser security:

   A. Configure browser's Privacy and Security settings.

   B. Keep your web browser fully patched and up to date.

   D. Delete browsing history.

E. Disable pop-up blocking.

Ans. D. Disable pop-up blocking.

5. Open Wi-Fi is more vulnerable to:

A. Eavesdropping

B. Misuse of network for distribution of malware etc.

C. Financial Frauds

D. All of the above

Ans. D. All of the above

6. Digital footprints are stored _____

A. Temporary

B. Permanently

C. Never

D. None of these

Ans. B. permanently